

In this paper, all problem in the Silverman will be solved in complete form.

#7, 1. a,

Let  $C: z(t) = e^{it}$  ( $-\pi \leq t \leq \pi$ ). Evaluate

$$\int_C z^2 dz = \int_{-\pi}^{\pi} e^{2it} \cdot i \cdot e^{it} dt = i \int_{-\pi}^{\pi} e^{3it} dt = i \cdot \left[ \frac{1}{3} i \cdot e^{3it} \right]_{-\pi}^{\pi} = 0$$

$$\int_C \frac{1}{z} dz = \int_{-\pi}^{\pi} \frac{1}{e^{it}} \cdot i \cdot e^{it} dt = 2\pi i.$$

$$\int_C \frac{1}{z^2} dz = \int_{-\pi}^{\pi} i \cdot \frac{1}{e^{2it}} dt = i \cdot \left[ +i \cdot e^{-it} \right]_{-\pi}^{\pi} = -1 \cdot 0 = 0.$$

$$\int_C \overline{z} dz = \int_{-\pi}^{\pi} [\cos t - i \sin t] \cdot i \cdot e^{it} dt = i \int_{-\pi}^{\pi} \cos^2 t + \sin^2 t dt = 2\pi i.$$

#8.1.2. a)

Let  $p(z)$  be a polynomial of degree  $n$ , then

$$\int_{|z|=2} \frac{p(z)}{(z-1)^{n+2}} dz = 0$$

Proof. By the Cauchy integral formula,

$$\int_{|z|=2} \frac{p(z)}{(z-1)^{n+2}} dz = p^{(n+1)}(1) \cdot \frac{(n+1)!}{2\pi i} = 0.$$

#8.1.2. b)

Given  $n, m \in \mathbb{N}$ , show that  $\int_C \frac{(n-1)! e^z}{(z-z_0)^n} dz = \int_C \frac{(m-1)! e^z}{(z-z_0)^m} dz$ , where  $C$  is along  $z_0$ .

Proof. By the Cauchy integral formula,

$$\int_C \frac{(n-1)! e^z}{(z-z_0)^n} dz = \frac{2\pi i}{(n-1)!} \cdot (n-1)! \cdot e^{z_0} = 2\pi i \cdot z_0.$$

Since the value is independent for  $n$ , so proved.

#8.1.3. Evaluate for  $C: |z|=3$ ,

$$\int_C \frac{e^z}{z-2} dz = 2\pi i \cdot e^2.$$

$$\int_C \frac{e^z \sin z}{(z-2)^2} dz = 2\pi i \cdot (e^2 \sin 2 + e^2 \cdot \cos 2).$$

# 8.1.4.

$$\int_{|z|=2} \frac{1}{z^2-1} dz = \frac{1}{2} \int_{|z|=2} \frac{1}{z-1} - \frac{1}{z+1} dz = 0$$

$$\int_{|z|=3} \frac{z^2+3z-1}{(z-1)(z+2)} dz = \frac{1}{3} \int_{|z|=3} \frac{z^2+3z-1}{z-1} - \frac{z^2+3z-1}{z+2} dz = \frac{1}{3} \cdot 2\pi i \cdot (6) = 4\pi i$$

# 8.1.6.

observe that: for  $|z|=1$ ,  $\operatorname{Re} z = \frac{z+\bar{z}}{2} = \frac{1}{2} \cdot (z + \frac{1}{z})$ .

$$\text{so } \int_{|z|=1} (\operatorname{Re} z)^2 dz = \frac{1}{4} \int_{|z|=1} \frac{(z^2+1)^2}{z^2} dz = \frac{1}{4} \cdot 2\pi i \cdot 8 = 4\pi i.$$

And, for  $|z-1|=1$ ,  $z=1+e^{i\theta}$  ( $-\pi \leq \theta \leq \pi$ ), so  $\bar{z}=1+e^{-i\theta}$ , therefore

$$\bar{z} = 1 + \frac{1}{z-1} = \frac{z}{z-1}.$$

$$\text{Therefore } \int_{|z-1|=1} (\bar{z})^2 dz = \int_{|z-1|=1} \frac{z^2}{(z-1)^2} dz = 2\pi i \cdot 2 = 4\pi i.$$

And, for  $|z-1|=1$ ,  $\operatorname{Im} z = \operatorname{Im}(z-1) = \frac{1}{2i} \cdot (z - \bar{z}) = \frac{1}{2i} \cdot \left( \frac{z^2-2z}{z-1} \right)$ , so

$$\int_{|z-1|=1} (\operatorname{Im} z)^2 dz = -\frac{1}{4} \cdot \int_{|z-1|=1} \frac{(z^2-2z)^2}{(z-1)^2} dz = -\frac{1}{4} \cdot 0 = 0.$$

#8.1.11. Evaluate the limits

$$\lim_{z \rightarrow 0} \frac{e^z - 1 - z}{z^2}$$

#8.1.12. If  $f$  is analytic at  $z = z_0$ , and  $|f^{(n)}(z_0)| \leq n^k$  for each  $n$ .

Then  $f$  is an entire.

Proof. Since  $f$  is analytic at  $z_0$ , all derivatives  $f^{(n)}(z_0)$  exist and by the assumption,

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} \cdot (z - z_0)^n$$

Converses for any  $z \in \mathbb{C}$ , so  $f$  is analytic.

(More specifically,

$$\limsup_{n \rightarrow \infty} \frac{(n+1)^k}{(n+1)!} \cdot \frac{n!}{n^k} = \limsup_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^k \cdot \frac{1}{n+1} = 0.$$

#8.1.13. Show that the following series are analytic.

$$f(z) = \sum_{n=1}^{\infty} \frac{1}{n^z} \quad (\operatorname{Re} z > 1), \quad f(z) = \sum_{n=1}^{\infty} \frac{1}{z^2 - n^2} \quad (z \neq \pm 1, \pm 2, \dots)$$

Proof. Recall that:  $n^z = e^{z \cdot \ln n}$

$$2^z = e^{z \cdot \ln 2}$$

Section. 8.2.

#8.2.1. If  $f(z) = \sum_{n=0}^{\infty} a_n z^n$  is analytic in  $|z| < R$ , and  $f(z)$  is real when  $-R < z < R$ . Show that all  $a_n$  are real, and  $f(\bar{z}) = \overline{f(z)}$ .

Proof. Let  $g(z) = \overline{f(\bar{z})}$ . So that

$$g(z) = \sum_{n=0}^{\infty} \overline{a_n} \cdot z^n$$

By the assumption,  $f(x) - g(x) = \sum_{n=0}^{\infty} (a_n - \overline{a_n}) \cdot x^n = 0$  for all  $x \in (-R, R)$ .

So  $a_n - \overline{a_n} = 0$  for all  $n$ .

#8.2.2. Let  $p(z) = a_n z^n + \dots + a_0$  ( $a_n \neq 0$ ).

Given  $\varepsilon > 0$ , there exists  $R > 0$  such that

$$r > R \Rightarrow (1 - \varepsilon) \cdot |a_n| \cdot r^n \leq |p(z)| \leq (1 + \varepsilon) |a_n| \cdot r^n \quad \text{for all } |z| = r.$$

#8.2.3.

If all the zeros of a polynomial  $P(z)$  have positive real part, then all the zeros of  $P'(z)$  have positive real part.

Proof. Denote that  $P(z) = a_n z^n + a_{n-1} z^{n-1} + \dots + a_0 = a_n (z - z_1) \dots (z - z_n)$ .

Then,  $P'(z) = a_n \cdot \left[ \sum_{i=1}^n \prod_{j \neq i} (z - z_j) \right]$ , so therefore

$$\frac{P'(z)}{P(z)} = \sum_{i=1}^n \frac{1}{z - z_i} \quad \text{for } z \neq z_1, \dots, z_n.$$

If  $\operatorname{Re} z \leq 0$ , then  $\operatorname{Re}(z - z_i) < 0$  for all  $1 \leq i \leq n$ , so  $P'(z)/P(z)$  cannot be zero, so  $P'(z)$  cannot be zero. Therefore, if  $P'(z) = 0$ ,  $\operatorname{Re} z > 0$ .

#8.2.4.

Let  $f$  be an entire satisfying  $|f(z)| \leq |e^z|$  for all  $z \in \mathbb{C}$ .

Show that  $f(z) = k e^z$ ,  $|k| \leq 1$ .

Proof. Let  $g(z) = f(z)/e^z$ . Since  $f$  is entire, so is  $g$ .

And,  $|g(z)| = |f(z)|/|e^z| \leq 1$ , so  $g(z)$  must be a constant, by the Liouville's Thm.

Now,  $f(z) = g(z) \cdot e^z$ , where  $g$  is a constant satisfying  $|g| \leq 1$ .

#8.2.6.

Let  $f$  be an entire s.t.  $|f'(z)| \leq |z|$  for all  $z \in \mathbb{C}$ .

Then  $f(z) = az^2 + b$ , where  $a, b \in \mathbb{C}$  with  $|a| \leq \frac{1}{2}$ .

Proof. Given  $z_0 \in \mathbb{C}$ , and  $R > 0$ ,  $|f'(z)| \leq |z| = |z - z_0 + z_0| \leq |z - z_0| + |z_0|$ , for  $|z - z_0| = R$ .

So,  $|f''(z_0)| \leq \frac{1!}{R} \cdot \sup_{|z - z_0| = R} |f'(z)| \leq \frac{R + |z_0|}{R} = 1 + \frac{|z_0|}{R} \rightarrow 1$  as  $R \rightarrow \infty$ .

and, for  $n \geq 3$ ,

$$|f^{(n)}(z_0)| \leq \frac{(n-1)!}{R^{n-1}} \cdot \sup |f'| \leq \frac{R + |z_0|}{R^{n-1}} \rightarrow 0 \text{ as } R \rightarrow \infty.$$

And,  $|f'(z_0)| \leq \frac{1}{2\pi} \int_{\mathbb{C}} \frac{|f(z)|}{z} dz \leq \frac{2\pi R}{2\pi} \rightarrow 0 \text{ as } R \rightarrow \infty,$

Sec 9.2.

#1. Suppose that  $f(z)$  has an iso. sing. at  $z_0$ .

If  $\lim_{z \rightarrow z_0} (z - z_0)^\alpha \cdot f(z) = M$  or  $\infty$ , where  $M \neq 0$ , then  $\alpha$  must be integer.

proof Suppose that  $\lim_{z \rightarrow z_0} (z - z_0)^\alpha \cdot f(z) = M$ . Then, given  $\epsilon > 0$ , there is  $\delta > 0$  s.t.  
 $0 < |z - z_0| < \delta \Rightarrow |(z - z_0)^\alpha \cdot f(z)| < |M| + \epsilon$ .

So, the  $(z - z_0)^\alpha \cdot f(z)$  is bounded in the punctured  $0 < |z - z_0| < \delta$

Then,  $(z - z_0)^\alpha \cdot f(z)$  has a Taylor Expression

$$(z - z_0)^\alpha f(z) = a_0 + a_1(z - z_0) + \frac{1}{2!} \cdot a_2(z - z_0)^2 + \dots$$

#2. If  $f$  is analytic in  $B_R(0) \setminus \{0\}$  and  $\lim_{z \rightarrow 0} |zf(z)| = 0$ .

Then the origin is removable.

proof. Since

